9 The function $\mathrm{f}(x) = \dfrac{\mathrm{e}^x}{1-\mathrm{e}^x}$ is defined on the domain $x \in \mathbb{R}$, $x \neq 0$.

(i) Find $\mathrm{f}^{-1}(x)$. **[3]**

(ii) Write down the range of $\mathrm{f}^{-1}(x)$. **[1]**